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ASSIGNMENT

910
100

Question-1

- Encrypt the message "DEFEND THE WALL" using cipher with key = 3
- Decrypt the message "KHOOR ZRUOG" using Caesar cipher with key = 3
- A caesar cipher is used to encrypt the word "SECRET" to "VHFUHW" Determine the key used, and decrypt "WKLL LV D WHVW"

Sol! a) Encrypt "DEFEND THE WALL" using key = 3

Plain text :- DEFEND THE WALL

shift

D	E	F	E	N	D	T	H	E	W	A	L	L
G	H	I	H	Q	G	W	K	H	Z	D	O	O

cipher text :- GHIHQG WKH ZDOO

b) cipher text :- KHOOR ZRUOG

Shift

K H O O R Z R U O G

H E L L O W O R L D

Plain text :- HELLO WORLD

c) Given,

SECRET → VH FUHW

→ Key = S → V = +3

Decrypt

W K L V L V D W H V W

T H I S I S A T E S T

→ THIS IS A TEST

Question 2

• using the Substitution key

A → Q, B → W, C → E, D → R, E → T, F → Y . . .

Z → M Encrypt the word "FIRE"

• Given the cipher text "USHHD QEB

NRFZH VOLTK" Find plain text.

Sol: Substitution key

a) Encrypt "FIRE"

Step 1 Replace

F	I	R	E
Y	O	K	T

→ Cipher text :- YOKT

b) Decrypt

→ Cipher text :- UJHHD QEB NRFZH
YOLTK

UJHHD → QUICK

NRFZH → BROWN QEB → THE

YOLTK → FOX

∴ THE QUICK BROWN FOX

3) a) Encrypt "ATTACKATPALACE" us[ing]
Keyword "LEMON".

Plain text :- ATTACK AT PALACE

Keyword :- LEMON LEMON LEMO

-> Adding letter values (mod 26)

-> Cipher text :- LXFOPVEF DNWEOS

b) Decrypt "LXFOPVERN HR" using
Keyword "LEMON".

Cipher text :- LXFOPVEFRNHR

Keyword :- LEMON LEMON LE

-> Subtract the key values (mod 26)

-> Plain text :- ATTACK AT DAWN

c) Decrypt "QNXEPKMAE ATU" using
Keyword "KEY".

-> Cipher text :- QNXEPKMAE

-> Keyword :- KEYKEYKEYKEY

-> Plain text :- AJZULMCWGWPW

4) a) Encrypt "HI" using the Hill cipher

Key Matrix, $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

→ convert into numbers

H	I
7	8

$\begin{bmatrix} 7 \\ 8 \end{bmatrix}$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$$

→ Take mod 26

45 mod 26 = 19 ; 54 mod 26 = 2

19 = T ; 2 = C

→ HI = TC

b) verify "PO" → "QC"

P = 15 O = 14

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 15 \\ 14 \end{bmatrix} = \begin{bmatrix} 87 \\ 100 \end{bmatrix}$$

→ 87 mod 26 = 9 = J

→ 100 mod 26 = 22 = W

∴ PO → JW

∴ Not Verified

5) a) Encrypt "MEET ME AFTER LUNCH" using "ZEBRAS".

→ Keyword :- Z E B R A S

M E E T M E

A F T E R L

U N C H X X

→ A B E R S Z

5 3 2 4 6 ~~1~~

→ MRC ETC TEH EXL MAU

(sorted order)

b) Decrypt "EANMTEFCUHETREL" using key word "BOARD".

→ Key word :- B O A R D

2 4 1 3 3

→ MEET AFTER LUNCH

6) Encrypt "HELLO" using OTP key "XMCKL".

letter	H	E	L	L	O
Value	7	4	11	11	14

~~Letter~~ Key	X	M	C	K	L
Value	23	12	2	10	11

-) Adding mod 26

-) EQNVZ

b) OTP unbreakable

Message :- ATTACK KNOW

Key :- QWERTYUIO

-) under ideal conditions, OTP is mathematically unbreakable because

- The key is random
- The key is used only once.

7) a) Encrypt "WE ARE DISCOVERED" using columnar Transposition (KW: ZEBRA)

→ WE ARE DISCOVERED

→

Z	E	B	R	A
W	E	A	R	E
D	I	S	C	O
V	E	R	E	D

→ Keyword : Z E B R A

order : 5 3 2 4 1

→ AO : A → B → E → R → Z

EOD ASR EIE RCE WDV

→ EODASREIE RCE WDV

b) Decrypt "EVLNEACDTKESEAOROFO JDEECUXWN") Keyword: TRIAL

→ Keyword = TRIAL

→ AO :- A I L R T

3 4 5 2 1

-> WE ARE DISCOVERED FLEE AT ONCE.

8) Frequency Analysis.

Most frequent! Z, O, P, U, S

English frequencies

E, T, A, O are most common letters

Replace

Z -> E

O -> T

P -> A

U -> O

S -> I

Plain text!- IT WAS A SECRET MESSAGE

9) Decrypt using known plaintext

-> known plain text!- THIS IS THE

B -> T

J -> H

Q -> J

W -> S

Y -> H

M -> E

-> THIS IS THE SIMPLEST.

10) Recover the key & Decrypt

-> Asume a short key
-> cipher text :- ZICVTWQNGRZGVTMAZH
-> Recovered key : KEY QCYGLMCJ"!

-> Decrypt using the key

Answer : KEY

-> plain text :- WE ARE DISCOVERED

SAVE YOURSELF